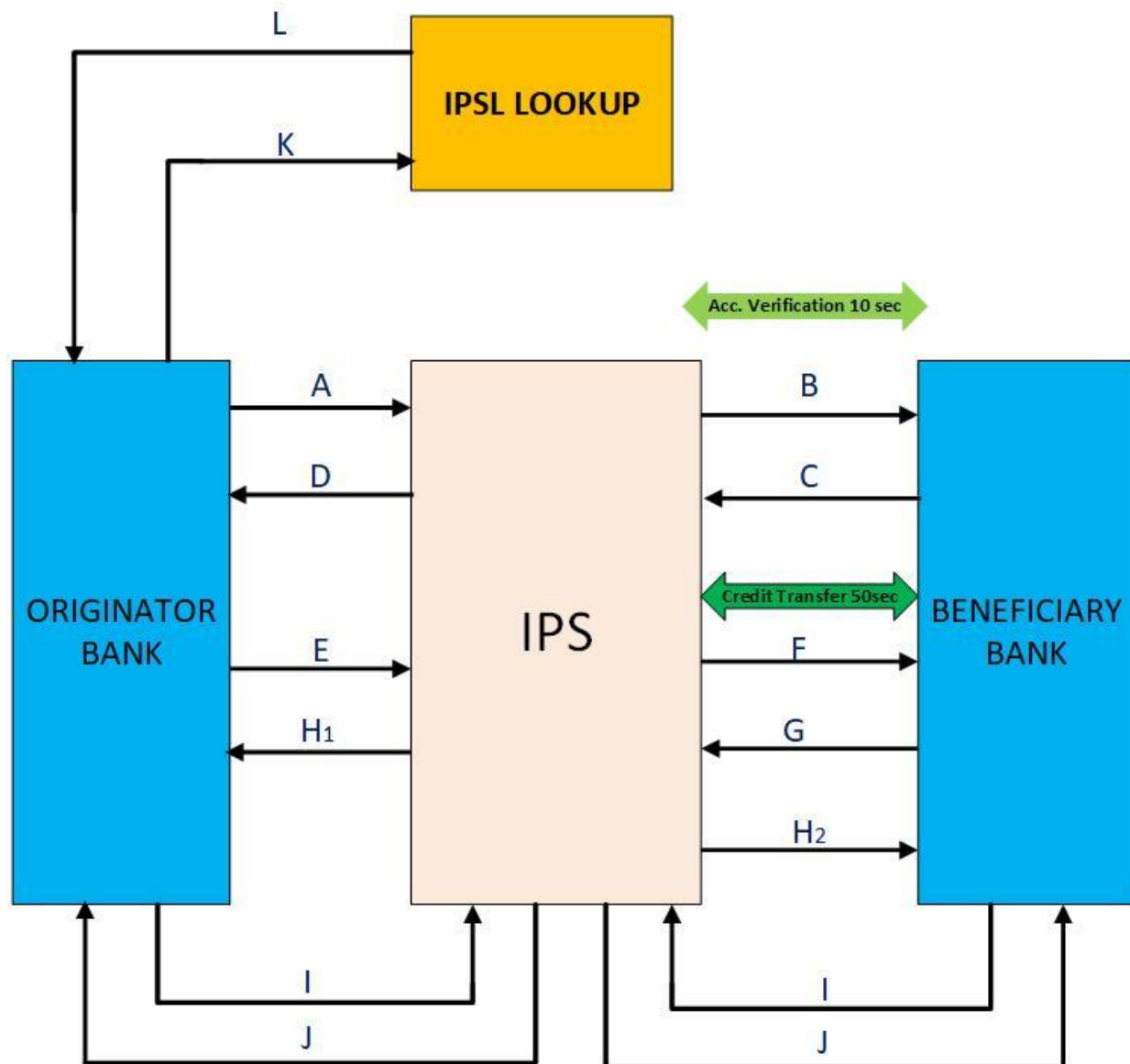


IPS TRANSACTION WORKFLOW

V 1.0

23rd Feb 2021

1. CREDIT TRANSFER



Account Verification (Send to Account)

Once a customer initiates an account-to-account transaction on their bank channel;

- A. The Originator Bank Initiates Account Verification Request containing beneficiary customer account number and beneficiary bank.
- B. IPS forwards the validated Account Verification Request to the Beneficiary Bank.

- C. Beneficiary bank performs payee's account number verification against the details on the core banking system and returns to IPS Account Verification Report containing verification results, which include the beneficiary name.
- D. IPS forwards the Account Verification Report to the Originator Bank, who confirms with sending customer, that they are paying to the correct recipient. Originator bank proceeds to initiate a Credit Transfer financial message.

Account Verification (Send to Phone)

Once a customer initiates an account-to-phone transaction on their bank channel;

- K. The Originator Bank Initiates Account Verification Request containing beneficiary customer phone number, to IPSL lookup.
- L. IPSL lookup responds with the beneficiary details including;
 - Banks that the beneficiary phone number is registered,
 - Name of beneficiary
 - Beneficiary account number.

Originator Bank, confirms with the sending customer, that they are paying to the correct recipient. Originator bank proceeds to initiate a Credit Transfer financial message.

NB: Originator bank to have an inbuilt mechanism for account validation request that does not result into credit transfer.

Credit Transfer

- E. The Originator Bank initiates a Credit Transfer message to the IPS. The IPS validates the message, by performing below tasks;
 - Checks status of the Originator Bank and Beneficiary Bank,
 - Checks existing limits for the participants
 - Checks whether the Originator Bank's net position allows performing the transaction.

- F. If all the validations performed are successful, IPS forwards the Credit Transfer message to the Beneficiary Bank and starts the timeout counter (Maximum of 50 seconds).
- G. Beneficiary Bank receives the credit transfer message. Beneficiary bank put a hold on account to be credited and responds with the positive confirmation, with the Credit Transfer Status Report message.
- H. IPS receives the response before timeout expires. It positively validates the received response Credit Transfer Status Report message and changes net positions on the participants.

H1. The IPS then submits Credit Transfer Status Report message to the Originator Bank. Originator bank informs the customer about the successful transaction.

H2. The IPS also submits Credit Transfer Status Report message to the Beneficiary Bank. Based on this message the Beneficiary Bank credits the beneficiary customers' account.

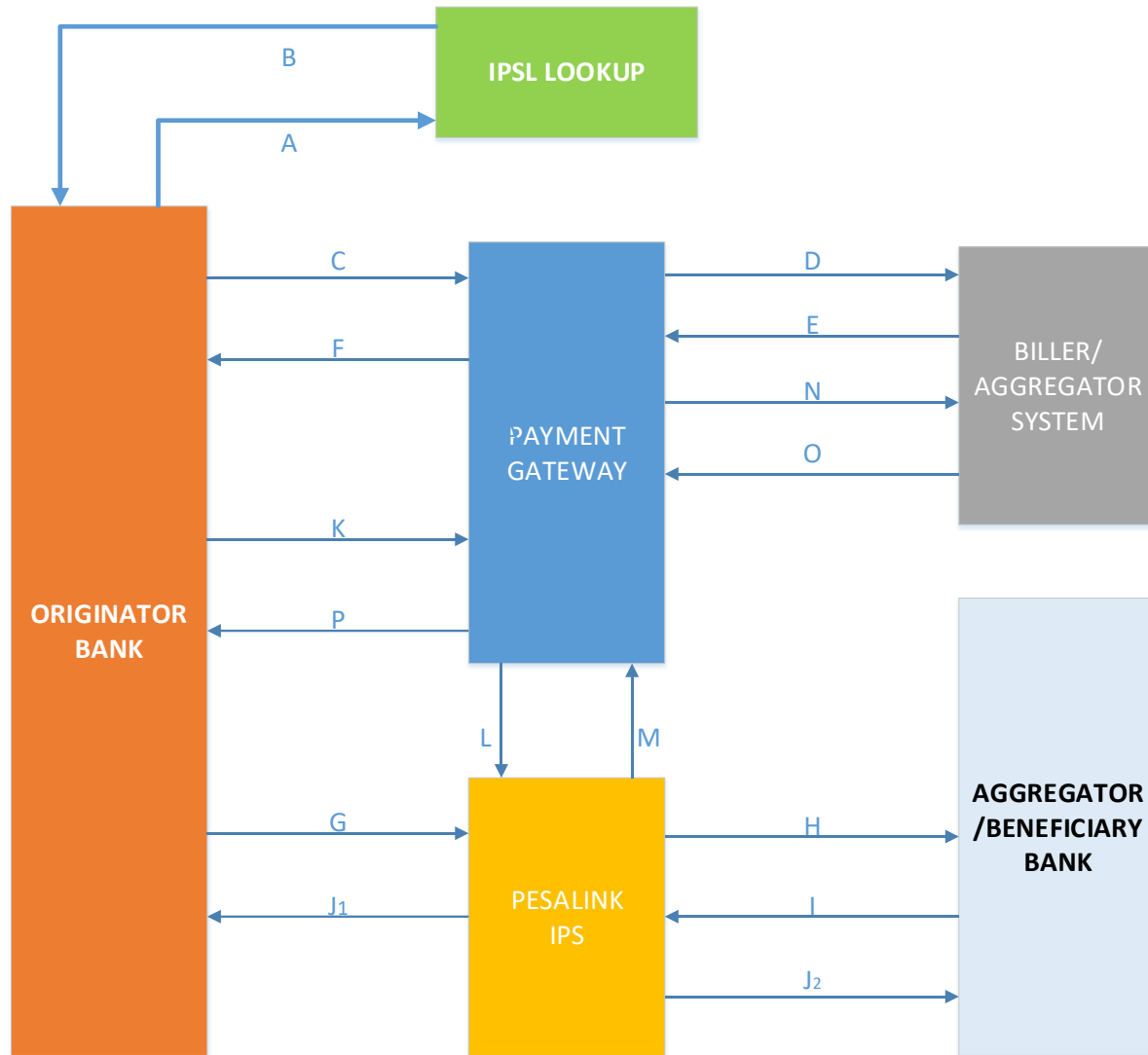
Investigative Message

If there is no final confirmation (Credit Transfer Status Report message) received from the IPS as result of Credit Transfer processing (H1), the Originator Bank can initiate an investigative message, before finalizing the debit transaction with its customer. Similarly, if the Credit Transfer Status Report (H2) was not delivered to Beneficiary Bank, the Beneficiary Bank must initiate an Investigation message to get status of the transaction before crediting the beneficiary account.

- I. The Originator/Beneficiary Bank Initiates Investigative message to the IPS.
- J. The IPS validates the Investigative message, and finds the initial Credit Transfer transaction. The IPS returns to the Originator/Beneficiary Bank, with the Credit Transfer Status Report message with the actual status of the initial Credit Transfer transaction.

2. PAY TO BUSINESS (PAYBILL)

FLOW DIAGRAM



Customer initiates a send to business transaction with Bill reference number and corresponding amount, as provided by the biller/aggregator;

Account Verification on MGW lookup

- A. The Originator Bank initiates Biller's Name Verification Request containing biller's paybill number, to IPSL lookup.

- B. IPSL lookup responds with the biller's details including;
- PIC code that the biller's paybill number is registered,
 - Name of biller
 - Biller's account number.

Originator Bank, confirms with the sending customer, that they are paying to the correct biller.

NB: Originator bank to have an inbuilt mechanism for account validation request that does not result into credit transfer.

Validation request and response

- C. Originator bank proceeds to initiate the validation request to the PGW, that contains bill reference/account number and amount.
- D. PGW forwards the amount and bill reference/account number to the biller for validation.
- E. Biller responds with positive confirmation if the validation was successful.
- F. PGW forwards the positive response to originator bank.

Credit Transfer

- G. The Originator Bank initiates a Credit Transfer message to the IPS. The IPS validates the message, by performing below tasks;
- Checks status of the Originator Bank and beneficiary Bank,
 - Checks existing limits for the participants,
 - Checks whether the Originator Bank's net position allows performing the transaction.
- H. If all the validations performed are successful, IPS forwards the Credit Transfer message to the Beneficiary Bank and starts the timeout counter (Maximum of 50 seconds).

- I. Beneficiary Bank (where the biller's account resides) receives the credit transfer message. Beneficiary bank put a hold on account to be credited and responds with the positive confirmation, with the Credit Transfer Status Report message
- J. IPS receives the response before timeout expires. It positively validates the received response Credit Transfer Status Report message and changes net positions on the participants.

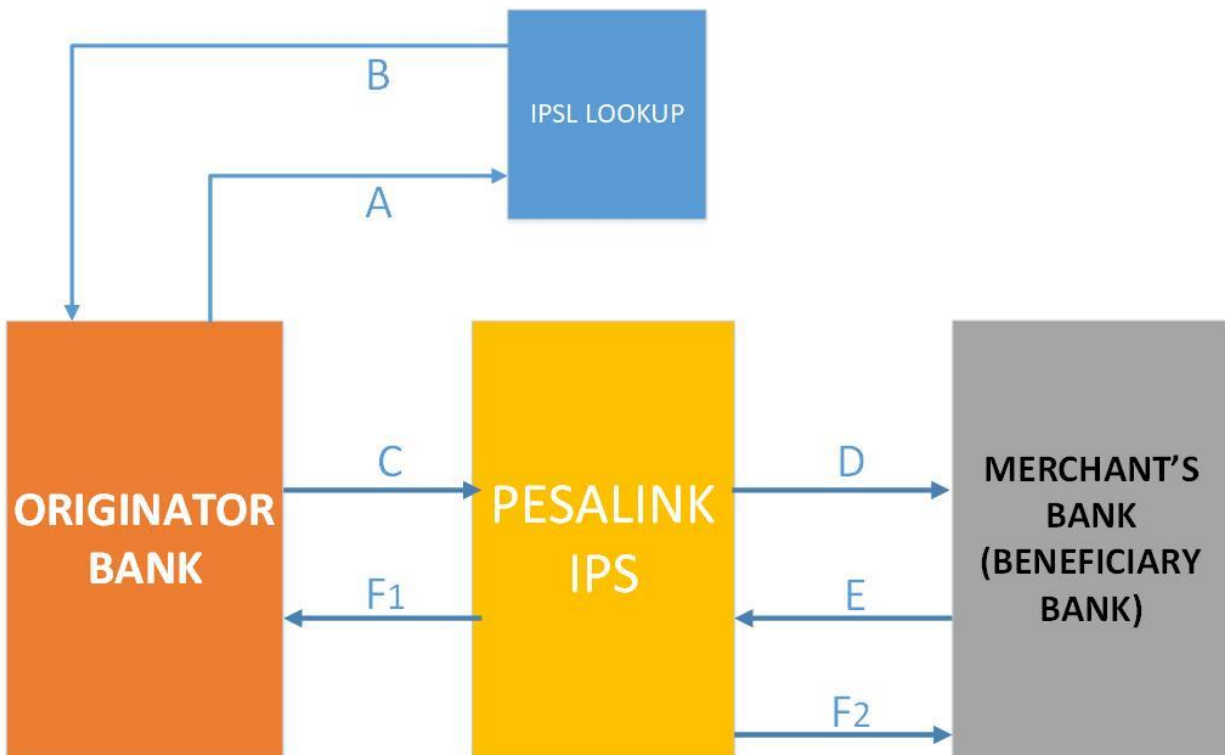
J1. The IPS then submits Credit Transfer Status Report message to the Originator Bank. Originator bank informs the customer about the successful transaction.

J2. The IPS also submits Credit Transfer Status Report message to the Beneficiary Bank. Based on this message the Beneficiary Bank credits the beneficiary customers' account.

Confirmation request and response

- K. After originator bank receives successful Credit Transfer Status Report message, originator bank initiates a payment confirmation request to the PGW.
- L. PGW forwards the request to IPS to confirm status of the transaction.
- M. IPS responds with status of the transaction to the PGW.
- N. PGW forwards the success status to the biller, for the biller to complete offering the service to the customer.
- O. Biller acknowledges receipt of funds into their account.
- P. PGW also responds to originator bank with success status.

3. PAY TO BUSINESS (MERCHANT)



Customer initiates a pay to till transaction with amount and merchant's till number

Account Verification on MGW lookup

- A. The Originator Bank initiates Merchant's Name Verification Request containing the merchant's till number, to IPSL lookup.
- B. IPSL lookup responds with the merchant's details including;
 - PIC code of the bank that the merchant's till is registered,
 - Name of merchant
 - Merchant's account number.

Originator Bank, confirms with the sending customer, that they are paying to the correct merchant.

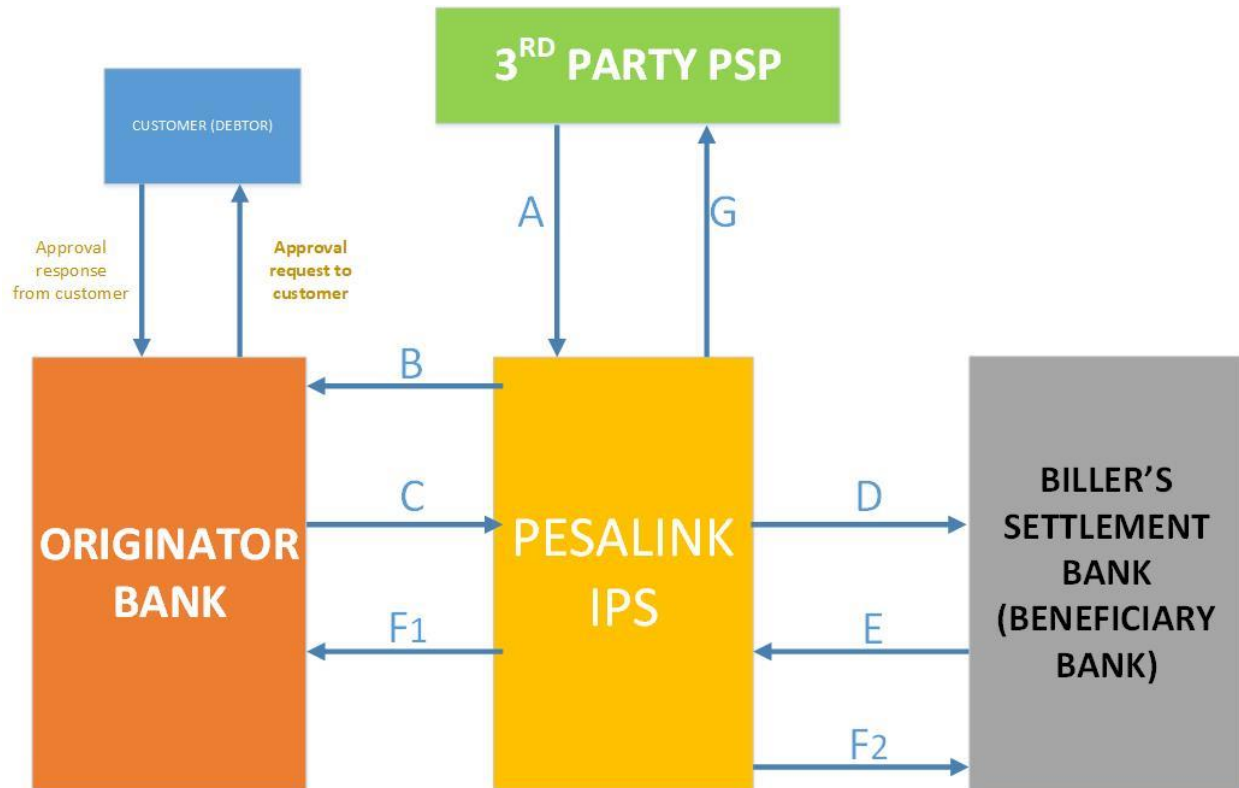
NB: Originator bank to have an inbuilt mechanism for account validation request that does not result into credit transfer.

Credit Transfer

- C. The Originator Bank initiates a Credit Transfer message to the IPS. The IPS validates the message, by performing below tasks;
- i. Checks status of the Originator Bank and beneficiary Bank,
 - ii. Checks existing limits for the participants,
 - iii. Checks whether the Originator Bank's net position allows performing the transaction.
- D. If all the validations performed are successful, IPS forwards the Credit Transfer message to the Beneficiary Bank and starts the timeout counter (Maximum of 50 seconds).
- E. Beneficiary Bank (where the merchant's account resides) receives the credit transfer message. Beneficiary bank put a hold on account to be credited and responds with the positive confirmation, with the Credit Transfer Status Report message
- F. IPS receives the response before timeout expires. It positively validates the received response Credit Transfer Status Report message and changes net positions on the participants.
- F1.** The IPS then submits Credit Transfer Status Report message to the Originator Bank. Originator bank informs the customer about the successful transaction.
- F2.** The IPS also submits Credit Transfer Status Report message to the Beneficiary Bank. Based on this message the Beneficiary Bank credits the merchant's account and notifies merchant about successful transaction.

4. REQUEST TO PAY

The IPS solution supports the ability to initiate a Credit Transfer request by Third-party



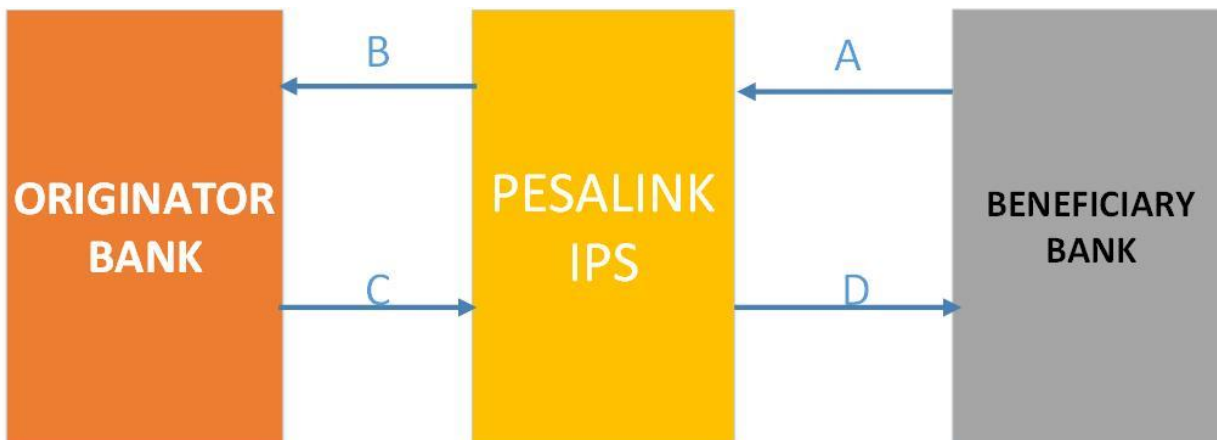
TRANSACTION FLOW (STEPS)

A. The Third-party PSP (Payment Service Provider) submits Customer Credit Transfer initiation message to the IPS. The message contains data necessary for payment initiation:

- Originator Bank name
- Beneficiary Bank name
- Ultimate Debtor's name and account number
- Ultimate Creditor's name and account number
- Payment details.

- B. The IPS system validates the payment initiation message and forwards Customer Credit Transfer initiation message to the Originator Bank.
- C. The Originator Bank has to communicate the request with its customer (Ultimate Debtor), and to receive approval for the Credit Transfer initiation.
If the Originator Bank's customer (Ultimate Debtor) confirms the payment, the Originator Bank debits the customer and initiates the Credit Transfer message.
- D. The IPS validates the message. If all the validations performed are successful, IPS forwards the Credit Transfer message to the Beneficiary Bank and starts the timeout counter (Maximum of 50 seconds).
- E. Beneficiary Bank (where the PSP's account resides) receives the credit transfer message. Beneficiary bank puts a hold on account to be credited and responds with the positive confirmation, with the Credit Transfer Status Report message
- F. IPS receives the response before timeout expires. It positively validates the received response Credit Transfer Status Report message and changes net positions on the participants.
 - F1.** The IPS then submits Credit Transfer Status Report message to the Originator Bank. Originator bank informs the customer about the successful transaction.
 - F2.** The IPS also submits Credit Transfer Status Report message to the Beneficiary Bank. Based on this message the Beneficiary Bank credits the PSP's account.
- G. After the Credit Transfer is successfully processed, the IPS initiates Payment Status Report message to inform the Third-party PSP about the final status of the initiated payment.

5. REQUEST TO PAY (WITH MANDATE)



The Payment Activation Request message is used as Request to Pay with Mandate ID message. The Request to Pay with Mandate ID message is initiated by the Beneficiary Bank to request an account to be debited at the receiving bank (the Originator Bank). The message contains Mandate related information, which is the authorization and expression of consent given by the Ultimate Debtor to the Ultimate Creditor to allow such debit on the debtor's account. The Mandate ID is generated during the pre-authorization procedure.

- A. The Beneficiary Bank (Initiating Participant) initiates Payment Activation Request to request the Originator Bank as the Request to Pay with Mandate ID message. The request message contains;
 - i. Originator Bank name
 - ii. Beneficiary Bank name
 - iii. Ultimate Debtor's name and account number
 - iv. Ultimate Creditor's name and account number
 - v. Payment details.
 - vi. Mandate ID (Mandate Code).
- B. The IPS successfully validates the received message and forwards the Payment Activation Request to the Originator Bank.
- C. Originator Bank performs necessary validation on the Mandate Identification (Mandate Code). Originator Bank confirms debit on the Ultimate Debtor's

account and responds with the Credit Transfer message. This response message contains the Mandate ID and reference ID of the initial message.

- D. The IPS validates the received message by performing below tasks;
- i. Checks status of the Originator Bank and beneficiary Bank,
 - ii. Checks existing limits for the participants,
 - iii. Checks whether the Originator Bank's net position allows performing the transaction.

If all the validations performed are successful, the IPS changes net positions on the participants. IPS then submits the final Credit Transfer Status Report message to the Beneficiary Bank, informing it of the positive result of initial Request to Pay with Mandate request.